Efficient Instrumental Variable Estimation under Sequential Missingness

Jiqing Zhao

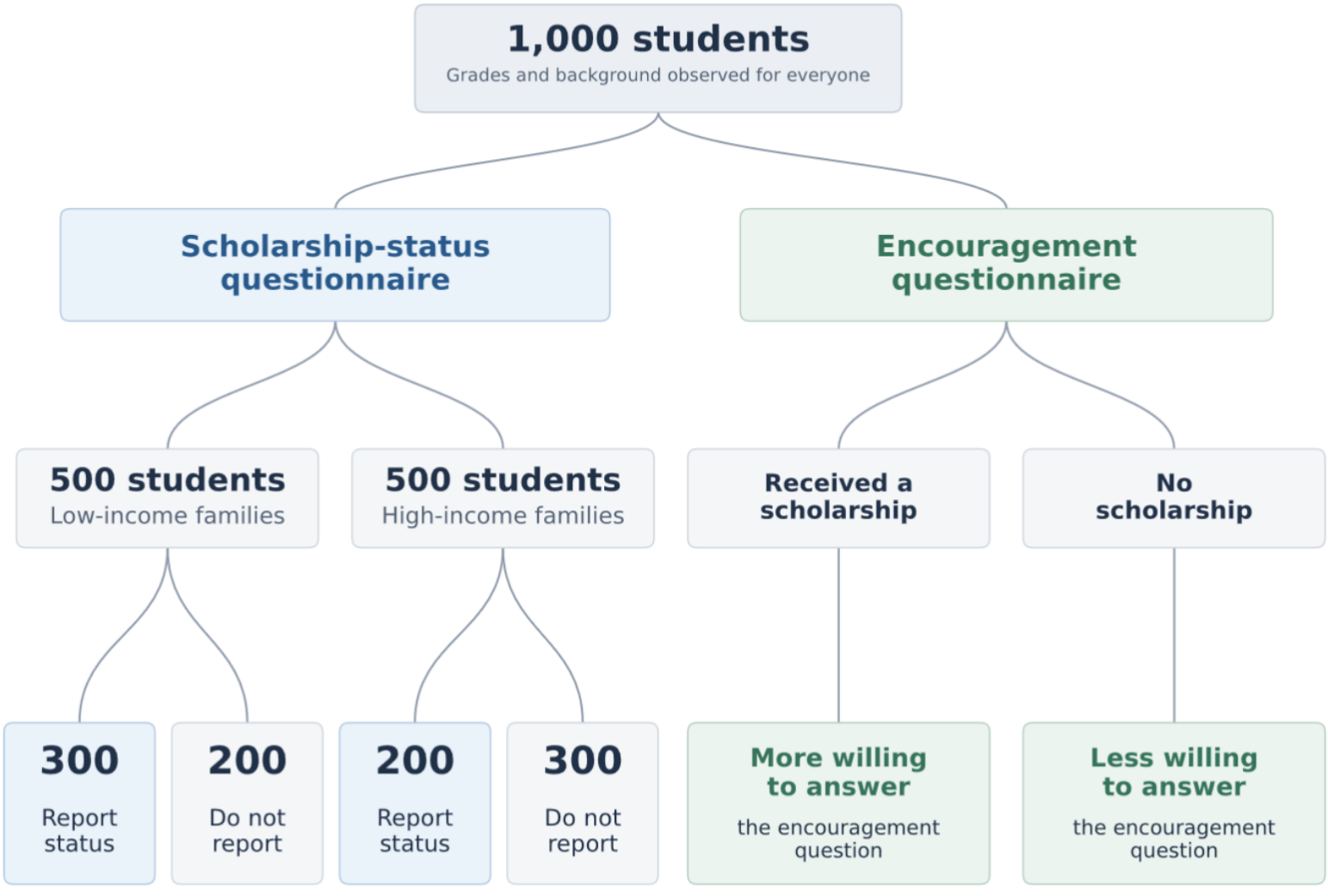
University of Texas at Austin
jiqingzhao@utexas.edu

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Motivation

- Simultaneous missingness in endogenous variables and instruments is common in empirical research
- I discuss recovery of available moments under the sequential missing at random (SMAR) assumption using an IPW-GMM approach
- SMAR allows instrument availability to depend on the endogenous variables when they are observed

Example - Missing Data in Scholarship and Grades Studies



Example - Information From Different Sources

- Different blocks come from survey modules, linked records, waves, or administrative sources
- Each block's data availability can depend on information already available at that stage
- SMAR is credible when such information accounts for selection into the next block

The Indonesian Family Life Survey (IFLS): child records, maternal records, and community facilities

Example - Family Size And Schooling

- Child records provide schooling and demographics
- Linked maternal records provide the number of siblings
- Community records provide family planning instruments

First layer

Demographics, community information, and schooling account for selection into a usable maternal record

Second layer

Once siblings are observed, the sibling count can help explain the available community-instrument pattern

IFLS: empirical design follows Chaudhuri and Guilkey (2016)

Literature

- Missing data and GMM approaches
  - **Foundations of missing-data analysis:** Dagenais (1973), Rubin (1976), Gourieroux and Monfort (1981), Little and Rubin (2002), Donders et al. (2006)
    $\Rightarrow$ Missingness mechanisms and linear model performance
    $\Rightarrow$ Simple fixes don't work well

  - **GMM approach:** Graham (2011), Abrevaya and Donald (2017), Coe (2018), Muris (2020), Yang (2026)
    $\Rightarrow$ GMM for missing covariates/instruments/unbalanced panels

- Imputation/G-computation
  - Azur et al. (2011), Stekhoven and Bühlmann (2012), Rai (2023), Robins (1986), Naimi et al. (2016), Chatton et al. (2020)

  - Distribution based/potential outcome
    $\Rightarrow$ GMM has a lighter modeling burden
    $\Rightarrow$ GMM can preserve IV structure

Road Map

- Model and Identification
  - Establish an identification strategy for IV models under SMAR
- Estimation
  - Propose a two-step IPW-GMM estimator
- Asymptotic Theory
  - Establish consistency, asymptotic normality, and optimal weighting
- Monte Carlo Simulations
  - Compare my estimator with other commonly used estimators
- Empirical Application
  - Apply my method to study the effect of the number of siblings in a family on a child's education (IFLS data)

Model

Consider a linear model:

$$Y = D'\alpha_0 + X'\beta_0 + \epsilon, \qquad E[\epsilon|X, Z] = 0$$

where

- Y is the dependent variable
- D is a vector of p endogenous variables, where $p > 0$
- X is a vector of q control variables, where $q \geqslant 0$
- ϵ is the error term

In addition, let Z denote the set of k ($k \geqslant p$) instruments

- D and Z can sometimes be missing
- ϵ satisfies $E(X\epsilon) = E(Z\epsilon) = 0$
- α_0 is the parameter vector of interest

Model

Additional notation:

- $\theta = (\alpha', \beta')' \in \Theta \subset \mathbb{R}^{p+q}$
- $V = (Y, X')'$ denotes the always-observed variables
- $W = (Y, X', D', Z')'$ denotes all variables
- $\tilde{\epsilon}(\theta) := Y - D'\alpha - X'\beta$
- $\tau(\theta) := \begin{pmatrix} Z\tilde{\epsilon}(\theta) \\ X\tilde{\epsilon}(\theta) \end{pmatrix} \in \mathbb{R}^{k+q}$
- $G := E[\partial\tau/\partial\theta']$ is the Jacobian

$\Rightarrow E[\tau(\theta)] = G(\theta - \theta_0)$ for every θ

Missingness Notation: 2-instrument Example

Suppose there is missing data in D and Z. Let $k = 2$

Let

- $g_D = 1$ if every element in D is observed, $g_D = 0$ otherwise
- $\lambda_Z \in \{s_1, \cdots, s_J\}$ denotes the instrument-observation pattern for Z
- $s_1 = \{1, 2\}$, $s_2 = \{1\}$, $s_3 = \{2\}$, $s_4 = \emptyset$
- $g_{Z,j} = \mathbb{1}\{\lambda_Z = s_j\}$
- $\tilde{S}_1\tau(\theta) = \begin{bmatrix} Z_1\tilde{\epsilon}(\theta) \\ Z_2\tilde{\epsilon}(\theta) \\ X\tilde{\epsilon}(\theta) \end{bmatrix}$, $\tilde{S}_2\tau(\theta) = \begin{bmatrix} Z_1\tilde{\epsilon}(\theta) \\ X\tilde{\epsilon}(\theta) \end{bmatrix}$, $\tilde{S}_3\tau(\theta) = \begin{bmatrix} Z_2\tilde{\epsilon}(\theta) \\ X\tilde{\epsilon}(\theta) \end{bmatrix}$, $\tilde{S}_4\tau(\theta) = \begin{bmatrix} X\tilde{\epsilon}(\theta) \end{bmatrix}$

▸ More

General Missingness Notation

Suppose there is missing data in D and Z

Let

- $g_D = 1$ if every element in D is observed, $g_D = 0$ otherwise
- $\lambda_Z \in \{s_1, \cdots, s_J\}$ denotes the instrument-observation pattern for Z
- $s_j \subseteq \{1, \cdots, k\}$
- $g_{Z,j} = \mathbb{1}\{\lambda_Z = s_j\}$
- $S_j \in \mathbb{R}^{k_j \times k}$ ($k_j := |s_j|$) be the selection matrix of instrument availability
- $\tilde{S}_j = \begin{bmatrix} S_j & 0 \\ 0 & I_q \end{bmatrix} \in \mathbb{R}^{(k_j+q)\times(k+q)}$

SMAR - Assumption

Assumption 1 (SMAR)

(i) $g_D \perp (D, Z)|V$
(ii) $\lambda_Z \perp Z|(D, V, g_D = 1)$

- Missingness can depend on always-observed variables
- SMAR allows missingness of instruments to depend on the observed part of partially observed variables
- No restriction on instrument availability when $g_D = 0$

SMAR - Example

Assumption 1 (SMAR)

(i) $g_D \perp (D, Z)|V$
(ii) $\lambda_Z \perp Z|(D, V, g_D = 1)$

- Y is a child's schooling
- D is the number of siblings of a child
- X is demographics or residential controls
- Z is family planning infrastructures
- g_D is whether the maternal record and child's record can be linked
- λ_Z is the patterns of infrastructure availability

SMAR - Nested Information

The SMAR selection follows

$$\sigma(V) \subseteq \sigma(D, V) \subseteq \sigma(W)$$

- V is available for everyone
- Observing D makes the structural residual computable
- Each observed instrument then provides a feasible IV moment

The conditioning information sets are nested

The instrument patterns can be nonmonotone

SMAR - Joint Probability

Define the two data availability probabilities and their product:

$$\begin{aligned} p_D(V) &= P(g_D = 1|V) \\ p_{Z,j}(D,V) &= P(\lambda_Z = s_j|D,V,g_D = 1) \\ p_j(D,V) &= p_D(V)p_{Z,j}(D,V) \end{aligned}$$

Under SMAR, the joint probability of data availability can be recovered from observed data:

$$\begin{aligned} & P(\lambda_Z = s_j, g_D = 1|V,Z,D) \\ =& P(\lambda_Z = s_j|V,Z,D,g_D = 1)P(g_D = 1|V,Z,D) \\ =& p_{Z,j}(D,V)p_D(V) \\ =& p_j(D,V) \end{aligned}$$

The order of SMAR matters and may change according to specific datasets

Moment Conditions

Let

$$w_j := \frac{g_D g_{Z,j}}{p_j(D, V)}$$

For stratum j, the IPW moment condition is

$$E[w_j \tilde{S}_j \tau(\theta_0)] = 0 \qquad j = 1, \cdots, J$$

Then, let $A_j \in \mathbb{R}^{(p+q)\times(k_j+q)}$ be a set of weighting matrices such that the weighted sum of moments forms a single system of dimension $p + q$:

$$m(\theta) := \sum_{j=1}^{J} A_j w_j \tilde{S}_j \tau(\theta)$$

In addition, let

$$\bar{A} := \sum_{j=1}^{J} A_j \tilde{S}_j$$

Identification - Assumptions

Suppose $(W_i, g_{D,i}, \lambda_{Z,i})_{i=1}^n$ is an i.i.d. sample

Let $(V_i, g_{D,i}, g_{D,i}D_i, g_{D,i}g_{Z,ij}, g_{D,i}S_{\lambda_{Z,i}}Z_i)_{i=1}^n$ denote the observed data

We impose the following additional assumptions for identification

Assumption 2 (Identification of the complete-data model)

$rank(G) = p + q$, equivalently $E[(Z', X')'(D', X')]$ has full column rank

Assumption 3 (Coverage)

For every instrument $\kappa \in \{1, \cdots, k\}$, there exists a stratum j with $\kappa \in s_j$ and $P(g_D = 1, \lambda_Z = s_j) > 0$

Identification - Assumptions

We impose the following additional assumptions for identification:

Assumption 4 (Overlap)

There exists $\delta \in (0,1)$ *such that* $p_D(V) \geqslant \delta$ *a.s. and* $p_{Z,j}(D,V) \geqslant \delta$ *a.s for every stratum with* $P(\lambda_Z = s_j | g_D = 1) > 0$

Assumption 5 (Moments)

$E\|t\|^2 < \infty$ *and* $E\|T\|^2 < \infty$, *where* $t := \begin{pmatrix} ZY \\ XY \end{pmatrix}$ *and*

$T := \begin{pmatrix} Z \\ X \end{pmatrix} \begin{pmatrix} D' & X' \end{pmatrix}$

Assumption 6 (Weighting matrices)

The collection $\{A_j\}_{j=1}^J$ *satisfies the condition that* $\bar{A}G = \sum_{j=1}^J A_j \tilde{S}_j G$ *is nonsingular*

Identification

Theorem 1 (Identification)

Under Assumptions 1-5:

(a) θ_0 *is the unique solution to the stacked moment conditions* $E[w_j \tilde{S}_j \tau(\theta)] = 0$ *for every* j *with* $P(g_D = 1, \lambda_Z = s_j) > 0$*;*

(b) *For any fixed matrices* $\{A_j\}_{j=1}^J$, $E[m(\theta)] = \bar{A}G(\theta - \theta_0)$ *for all* θ*. Hence,* $E[m(\theta)] = 0$ *has* θ_0 *as its unique solution if* $\bar{A}G$ *is nonsingular (Assumption 6).*

Similar to Muris (2020), in an extreme case where we never observe all instruments at once and no single stratum identifies the model on its own, combining the strata can identify θ_0

Estimation

We propose a two-step IPW-GMM estimator for this model:

Step 1 (propensity score):

- *Estimate $p_D(\cdot)$ by a nonparametric (e.g. series or other sieve estimators) regression of g_D on V using all observations*
- *Estimate $p_{Z,j}(\cdot)$ by a nonparametric multinomial regression of the vector of stratum indicators $(g_{Z,1}, \cdots, g_{Z,J})$ on (D, V) using the subsample with $g_D = 1$*
- *Trim all estimated probabilities to $[\tilde{\delta}, 1]$ for a small $\tilde{\delta} \in (0, \delta]$*
- *Obtain*

$$\hat{p}_j = \hat{p}_D \hat{p}_{Z,j}$$

$$\hat{w}_{ij} = \frac{g_{D,i} g_{Z,ij}}{\hat{p}_j(D_i, V_i)}$$

Estimation

We propose a two-step IPW-GMM estimator for this model:

Step 2 (GMM):

Solve

$$\hat{g}_n(\theta) := \frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{J} A_j \hat{w}_{ij} \tilde{S}_j \tau_i(\theta) = 0$$

and denote the solution by $\hat{\theta}$

▸ More on consistency

- Since the IV model is linear, we have a closed-form solution

▸ More

Asymptotics - Notation

Define

$$\mu_1(D, V; \theta) := E[\tau(\theta)|D, V] = \begin{pmatrix} m_Z(D, V)\tilde{\epsilon}(\theta) \\ X\tilde{\epsilon}(\theta) \end{pmatrix}$$

$$m_Z(D, V) := E[Z|D, V]$$

$$\mu_0(V; \theta) := E[\mu_1(D, V; \theta)|V]$$

$$e_D := \mu_1 - \mu_0$$

$$e_Z := \tau - \mu_1 = \begin{pmatrix} (Z - m_Z)\tilde{\epsilon}(\theta) \\ 0_q \end{pmatrix}$$

Lemma

Under Assumption 4, for every j

$$w_j \tilde{S}_j \tau = \underbrace{\tilde{S}_j \mu_0 + \frac{g_D}{p_D} \tilde{S}_j e_D + w_j \tilde{S}_j e_Z}_{=:\ \psi_j^*} + \underbrace{\frac{g_D - p_D}{p_D} \tilde{S}_j \mu_0}_{=:\ r_{D,j}} + \underbrace{\frac{g_D}{p_D} \cdot \frac{g_{Z,j} - p_{Z,j}}{p_{Z,j}} \tilde{S}_j \mu_1}_{=:\ r_{Z,j}}$$

Asymptotic Distribution

Theorem 4 (Asymptotic distribution with estimated probabilities)

Under Assumptions 1-6 and standard first-step estimation assumptions

$$\sqrt{n}(\hat{\theta} - \theta_0) = -(\bar{A}G)^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \psi_i^* + o_p(1) \to^d N(0, \Gamma^*)$$

where, with all objects evaluated at θ_0,

$$\psi^* = \bar{A}\mu_0 + \frac{g_D}{p_D} \bar{A} e_D + \sum_{j=1}^{J} w_j A_j \tilde{S}_j e_Z$$

$$\Gamma^* = (\bar{A}G)^{-1} \Omega^* (\bar{A}G)^{-1'}$$

$$\Omega^* = \bar{A} M_0 \bar{A}' + \bar{A} M_D \bar{A}' + \sum_{j=1}^{J} A_j \tilde{S}_j M_{Z,j} \tilde{S}_j' A_j'$$

where $M_0 = E[\mu_0 \mu_0']$, $M_D = E[p_D^{-1} e_D e_D']$, and $M_{Z,j} = E[(p_j)^{-1} e_Z e_Z']$.

▸ More on first-step assumptions

Asymptotics - Efficiency Gain

Proposition 1 (Efficiency gain from estimated weights)

Under the conditions of Theorems 3 and 4,

$$\begin{aligned}\Omega_{known} - \Omega^* &= E[\frac{1-p_D}{p_D}\bar{A}\mu_0\mu_0'\bar{A}'] \\ &\quad + \sum_{j=1}^{J} E[\frac{1-p_{Z,j}}{p_D p_{Z,j}} A_j \tilde{S}_j \mu_1 \mu_1' \tilde{S}_j' A_j'] \\ &\quad - \sum_{j \neq l} E[\frac{1}{p_D} A_j \tilde{S}_j \mu_1 \mu_1' \tilde{S}_l' A_l'] \\ &\succeq 0\end{aligned}$$

Consequently,

$$(\bar{A}G)^{-1}\Omega_{known}(\bar{A}G)^{-1'} - \Gamma^* \succeq 0$$

(See Robins, Rotnitzky, and Zhao, 1994; Hirano, Imbens, and Ridder, 2003; Wooldridge, 2007; Prokhorov and Schmidt, 2009)

Inference

- Asymptotic variance estimator and its consistency ▸ More
- Optimal weighting ▸ More
- Optimal feasible estimator

Optimal Weighting

Theorem 5 (Optimal weighting for the observable moment conditions)

Under Assumptions 1-6, 8 and 9, define the GMM information matrix

$$\mathcal{I}_0 = G' \left[\sum_{j=1}^{J} \sum_{l=1}^{J} \tilde{S}_j' \left[(\Sigma^*)^+ \right]_{jl} \tilde{S}_l \right] G$$

The matrix $\mathcal{I}_0$ is positive definite. Its inverse

$$\Gamma_{eff} = \mathcal{I}_0^{-1}$$

is the lower bound on the limiting covariance of $\sqrt{n}(\hat{\theta} - \theta_0)$ among the proposed IPW-GMM estimators satisfying Assumption 8, over all weighting matrices satisfying Assumption 6.

Optimal Weighting

Under the conditions of Theorem 5, define

$$A_j^* = G' \sum_{l=1}^{J} \tilde{S}_l' [(\Sigma^*)^+]_{lj}, \qquad j = 1, \cdots, J$$

and

$$\bar{A}^* = \sum_{j=1}^{J} A_j^* \tilde{S}_j$$

Then

$$\bar{A}^* G = \mathcal{I}_0, \qquad \Omega^*(A_1^*, \cdots, A_J^*) = \mathcal{I}_0$$

The estimator using these weights satisfies

$$\sqrt{n}(\hat{\theta}^* - \theta_0) = -\mathcal{I}_0^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \sum_{j=1}^{J} A_j^* \psi_{ij}^* + o_p(1) \to^d N(0, \mathcal{I}_0^{-1})$$

Inference

We can establish feasible weights and inference from the available data:

- The IPW-GMM estimator with optimal weights is

$$\hat{\theta}^* = \left\{ \frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{J} \hat{A}_j^* \hat{w}_{ij} \tilde{S}_j T_i \right\}^{-1} \left\{ \frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{J} \hat{A}_j^* \hat{w}_{ij} \tilde{S}_j t_i \right\}$$

  where $t := \begin{pmatrix} ZY \\ XY \end{pmatrix}$, and $T := \begin{pmatrix} Z \\ X \end{pmatrix} \begin{pmatrix} D' & X' \end{pmatrix}$

▸ More

Monte Carlo - DGP

The DGP is as follows:

$$
\begin{aligned}
&X, Z_1, Z_2 \overset{\text{i.i.d.}}{\sim} N(0,1), \qquad (X, Z_1, Z_2) \perp (\eta, \epsilon), \\
&D = 0.1 + 0.3Z_1 + 0.1Z_2 + \eta, \\
&Y = 0.3D + 0.5X + \epsilon, \\
&(\eta, \epsilon)' \sim N\left((0,0)', \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}\right).
\end{aligned}
$$

The parameter of interest is $\alpha_0 = 0.3$

Monte Carlo - Missingness Design

Let

$$\pi_c(u) := c + (1 - 2c)\Phi(u), \qquad c = 0.05$$

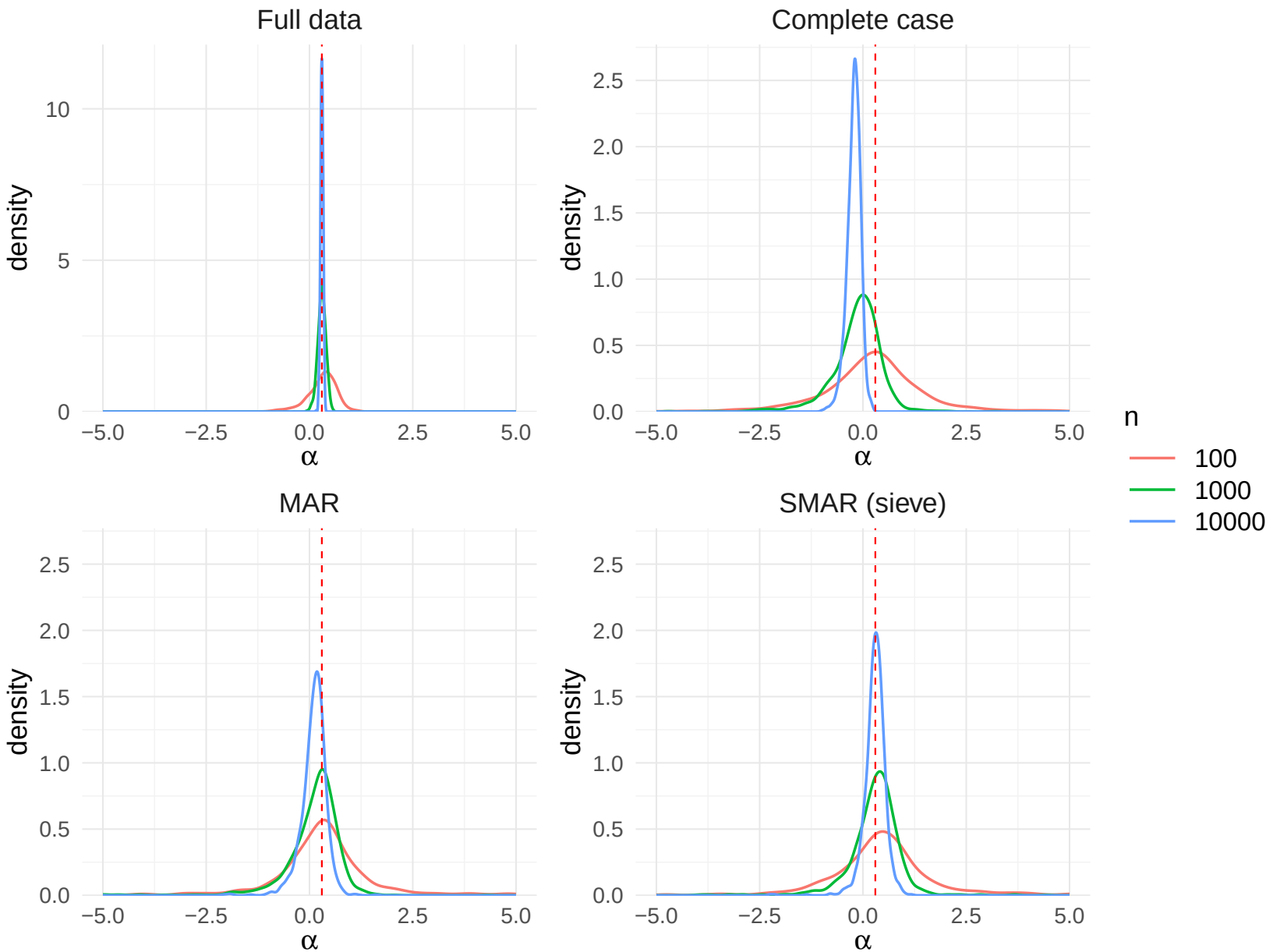
where Φ is the standard normal CDF. Set:

$$\begin{aligned} P(g_D = 0 \mid V) &= \pi_c(-0.6 + 0.2Y + 0.5Y^2), \\ P(\lambda_{Z_1} = 0 \mid D, V, g_D = 1) &= \pi_c(-0.4 - 0.1Y + 0.8D), \\ P(\lambda_{Z_2} = 0 \mid D, V, g_D = 1) &= \pi_c(-0.5 - 0.3Y - 0.5D + 0.8D^2). \end{aligned}$$

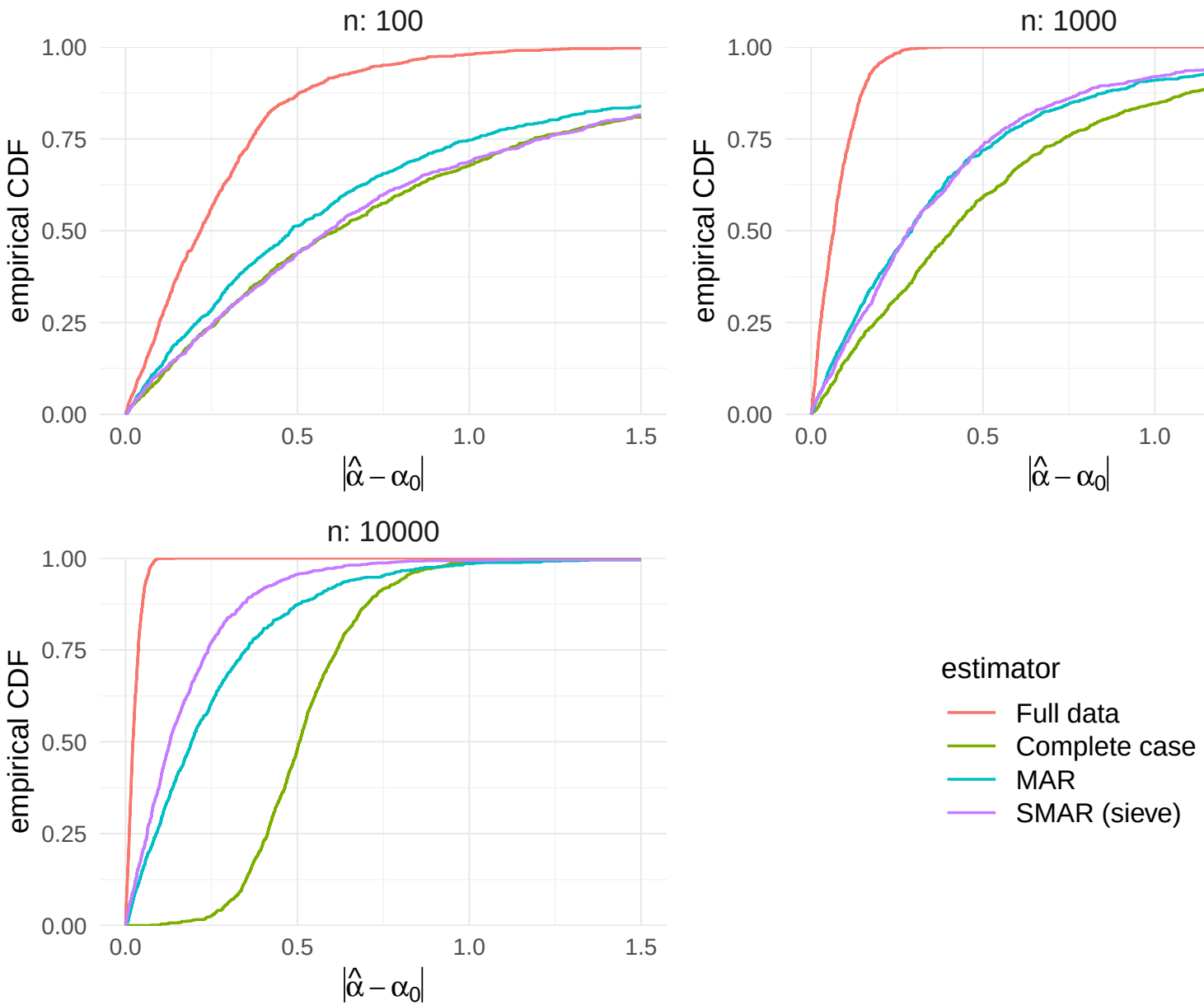
Note here the two instrument availability indicators are generated independently conditional on $(D, V, g_D = 1)$

Approximately 51% of observations have missing D, and only 14% are complete cases

Monte Carlo - Results



Monte Carlo - Results



Empirical Application - Introduction

- Does a larger number of siblings affect a child's schooling?
- Fertility and investment per child depend on the unobservables
- Family planning infrastructure provides instruments for fertility
- I extend the Chaudhuri and Guilkey (2016) design to include children with missing sibling counts

Empirical Application - Data

- IFLS waves used: 1993, 1997, 2000, and 2007
- Children ages 11-20, retained at their last eligible interview
- Raw household records linked to the CG community panel

Main analysis sample	16,238
Sibling count observed	10,817
All six instruments and sibling count observed	5,137

Empirical Application - Variables

- Y: years of schooling completed by the last eligible interview
- D: number of siblings, censored at four
- Child controls: age indicators, sex, and Muslim indicator
- Community controls: urban status and school counts
- Instruments: family planning posts, polindes, posyandu, village midwife, delivery post, village medicine post

Some controls are partially observed and require an extension of the always-observed X specification

Empirical Application - Instruments

Availability	Community family planning variables
Sometimes missing	Number of family planning posts
	Polindes indicator
Always observed	Number of posyandu
	Village midwife indicator
	Delivery post indicator
	Village medicine post indicator

Relevance: these facilities can affect the cost of fertility

Exclusion: conditional on controls, effects on schooling operate through family size

The exclusion restriction is inherited from CG

Empirical Application - Missingness

First layer: sibling counts require a usable mother–child link and maternal information

Second layer: conditional on $g_D = 1$,

Family planning posts	Polindes	Children
Observed	Observed	5,137
Observed	Missing	1,430
Missing	Observed	3,355
Missing	Missing	895

Each cell also contains the four always-observed instruments

Why SMAR?

First layer

Demographics, community information, and schooling account for selection into a usable maternal record

- Age and demographics can predict living arrangements and maternal co-residence
- Community characteristics can proxy the environment in which household information is collected
- Schooling can carry information about household circumstances beyond demographics

Second layer

Once siblings are observed, the sibling count can help explain the available community-instrument pattern

- Larger families may be more heavily represented in communities for which only one of the two instruments is recorded

Empirical Application: Main Results

Estimator	Estimate	SE
OLS (observed sibling count)	−0.076	0.016
GMM with four always-observed IVs	−1.113	0.297
Complete-case GMM with six IVs	−0.324	0.634
SMAR IPW-GMM	**−0.920**	**0.384**
CG (2016), AIPW-GMM, original sample	**−0.665**	**0.294**

- Trim 0.01 / 0.02 / 0.05: estimates −0.918 / −0.920 / −1.022
- Censoring the sibling count at five: estimate −0.756 (SE 0.329)

Extension

- Outcome Y is missing
- The order of SMAR changes

Conclusion

- An IPW-GMM approach for linear IV models with multivariate missingness under SMAR
- Asymptotic theory provides inference and optimal GMM weighting
- Monte Carlo simulations illustrate gains from accounting for the sequential missingness structure
- In the IFLS application, evidence shows that number of siblings have a negative effect on a child's schooling

Missingness Notation: 2-instrument Example

Let $q = 1$

- $S_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $S_2 = \begin{bmatrix} 1 & 0 \end{bmatrix}$, $S_3 = \begin{bmatrix} 0 & 1 \end{bmatrix}$, $S_4 = (\ ) \in \mathbb{R}^{0 \times 2}$
- $\tilde{S}_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $\tilde{S}_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $\tilde{S}_3 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $\tilde{S}_4 = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$
- $\tilde{S}_1\tau(\theta) = \begin{bmatrix} Z_1\tilde{\epsilon}(\theta) \\ Z_2\tilde{\epsilon}(\theta) \\ X\tilde{\epsilon}(\theta) \end{bmatrix}$, $\tilde{S}_2\tau(\theta) = \begin{bmatrix} Z_1\tilde{\epsilon}(\theta) \\ X\tilde{\epsilon}(\theta) \end{bmatrix}$, $\tilde{S}_3\tau(\theta) = \begin{bmatrix} Z_2\tilde{\epsilon}(\theta) \\ X\tilde{\epsilon}(\theta) \end{bmatrix}$, $\tilde{S}_4\tau(\theta) = \begin{bmatrix} X\tilde{\epsilon}(\theta) \end{bmatrix}$

▸ Back

Consistency

Assumption 7 (First-step conditions for consistency)

The estimated probabilities are trimmed to $[\tilde{\delta}, 1]$ *for some fixed* $\tilde{\delta} \in (0, \delta]$, *and*

$$\|\hat{p}_D - p_D\|_\infty = o_p(1), \qquad \max_{1 \leqslant j \leqslant J} \|\hat{p}_{Z,j} - p_{Z,j}\|_\infty = o_p(1)$$

Theorem 2 (Consistency)

Suppose Assumptions 1-7 hold. Then

$$\hat{\theta} \to_p \theta_0$$

▸ Back

Closed-form Solution

Closed-form Solution of the IPW-GMM Estimator

The solution of the GMM function is

$$\hat{\theta} = \hat{Q}_n^{-1} \frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{J} A_j \hat{w}_{ij} \tilde{S}_j t_i$$

where $\hat{Q}_n$ is assumed to be nonsingular and takes the form:

$$\hat{Q}_n = \frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{J} A_j \hat{w}_{ij} \tilde{S}_j T_i$$

where $t := \begin{pmatrix} ZY \\ XY \end{pmatrix}$ and $T := \begin{pmatrix} Z \\ X \end{pmatrix} \begin{pmatrix} D' & X' \end{pmatrix}$

▸ Back

Asymptotics - Assumptions

Assumption 8 (First-step estimation)

(i) *The estimated probabilities take values in* $[\tilde{\delta}, 1]$, $\tilde{\delta} \in (0, \delta]$

(ii) $\|\hat{p}_D - p_D\|_{L^2(V)} = o_p(n^{-1/4})$ *and*
$\max_j \|\hat{p}_{Z,j} - p_{Z,j}\|_{L^2(D,V|g_D=1)} = o_p(n^{-1/4})$

(iii) *The estimated probabilities are in a fixed Donsker class with probability approaching 1, and are uniformly consistent*

(iv) $E[\|\tau(\theta_0)\|^2] < \infty$

(v) *For every* j*:*

$$\frac{1}{n}\sum_{i=1}^{n} \hat{w}_{ij}\tilde{S}_j\tau_i(\theta_0) = \frac{1}{n}\sum_{i=1}^{n} \psi_{ij}^* + o_p(n^{-1/2})$$

Back

Asymptotic Variance Estimator

- Estimate the set of nuisance functions $\mu_{1,i}$ and $\mu_{0,i}$
- Define

  $\hat{e}_{D,i} = \hat{\mu}_{1,i} - \hat{\mu}_{0,i}, \qquad \tilde{S}_j \hat{e}_{Z,i} = \tilde{S}_j \tau_i(\hat{\theta}) - \tilde{S}_j \hat{\mu}_{1,i} \qquad \text{when } g_{D,i} g_{Z,ij} = 1$

- Estimate

$$\hat{M}_0 = \frac{1}{n} \sum_{i=1}^{n} \hat{\mu}_{0,i} \hat{\mu}_{0,i}', \quad \hat{M}_D = \frac{1}{n} \sum_{i=1}^{n} \frac{g_{D,i} \hat{e}_{D,i} \hat{e}_{D,i}'}{\hat{p}_D(V_i)^2}$$

  and estimate the observable instrument block by:

$$\widehat{\tilde{S}_j M_{Z,j} \tilde{S}_j'} := \frac{1}{n} \sum_{i=1}^{n} \frac{g_{D,i} g_{Z,ij}}{\hat{p}_j(D_i, V_i)^2} (\tilde{S}_j \hat{e}_{Z,i})(\tilde{S}_j \hat{e}_{Z,i})'$$

▸ Back

Asymptotic Variance Estimator

- Finally, obtain

$$\hat{\Omega}^* = \bar{A}(\hat{M}_0 + \hat{M}_D)\bar{A}' + \sum_{j=1}^{J} A_j \tilde{S}_j \widehat{M_{Z,j}} \tilde{S}_j' A_j'$$

$$\hat{\Gamma}^* = \hat{Q}_n^{-1} \hat{\Omega}^* \hat{Q}_n^{-1'}$$

where

$$\hat{Q}_n := \frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{J} A_j \hat{w}_{ij} \tilde{S}_j \begin{pmatrix} Z_i \\ X_i \end{pmatrix} \begin{pmatrix} D_i' & X_i' \end{pmatrix}$$

▸ Back

Asymptotic Variance Estimator - Consistency

Proposition 2 (Consistency of the variance estimator)

Under the conditions of Theorem 4, if

$$\frac{1}{n}\sum_{i=1}^{n}\|\hat{\mu}_{0,i}-\mu_0(V_i)\|^2 = o_p(1)$$

$$\frac{1}{n}\sum_{i=1}^{n} g_{D,i}\|\hat{\mu}_{1,i}-\mu_1(D_i,V_i)\|^2 = o_p(1)$$

then

$$\hat{\Omega}^* \to_p \Omega^*, \qquad \hat{\Gamma}^* \to_p \Gamma^*$$

▸ Back

Optimal Weighting - Notation

Consider the covariance relevant for optimal weighting:

$$\Sigma^* := Var \begin{pmatrix} \psi_1^* \\ \vdots \\ \psi_J^* \end{pmatrix}$$

For the (j, l) block:

$$\Sigma_{jl}^* = \tilde{S}_j(M_0 + M_D)\tilde{S}_l' + \mathbb{1}\{j = l\}\tilde{S}_j M_{Z,j}\tilde{S}_j'$$

And consequently

$$\Omega^*(A_1, \cdots, A_J) = \sum_{j=1}^{J}\sum_{l=1}^{J} A_j \Sigma_{jl}^* A_l'$$

Although the strata are disjoint, ψ_j^* share variation through μ_0 and e_D. Their covariance need not vanish ▸ Back

Optimal Weighting - Assumptions

Assumption 9 (Nondegeneracy)

(i) θ_0 *is an interior point of* Θ

(ii) *After removing redundant moment conditions, the covariance matrix of the remaining adjusted moments is positive definite, i.e.*

$$\Sigma_{reduced} \succ 0$$

(i) is standard; and (ii) states every nonzero remaining linear combination has sampling variation ▸ Back

Inference

We can establish feasible weights and inference from the available data:

- Let $\hat{G}$ be a consistent estimate of G and the covariance blocks be

$$\hat{\Sigma}^*_{jl} = \tilde{S}_j(\hat{M}_0 + \hat{M}_D)\tilde{S}_l' + \mathbb{1}\{j = l\}\widehat{\tilde{S}_j M_{Z,j} \tilde{S}_j'}$$

- Construct the feasible weights after removing redundant adjusted moments

$$\hat{A}^*_j = \hat{G}' \sum_{l=1}^{J} \tilde{S}_l' [(\hat{\Sigma}^*)^+]_{lj} \to_p A^*_j$$

- The IPW-GMM estimator with optimal weights is

$$\hat{\theta}^* = \left\{ \frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{J} \hat{A}^*_j \hat{w}_{ij} \tilde{S}_j T_i \right\}^{-1} \left\{ \frac{1}{n} \sum_{i=1}^{n} \sum_{j=1}^{J} \hat{A}^*_j \hat{w}_{ij} \tilde{S}_j t_i \right\}$$

where $t := \begin{pmatrix} ZY \\ XY \end{pmatrix}$, and $T := \begin{pmatrix} Z \\ X \end{pmatrix} \begin{pmatrix} D' & X' \end{pmatrix}$

▸ Back